

Yifan Zhang

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Research Interest

I am interested in research that emphasizes rigorous empirical evaluation, so that the main findings remain reliable beyond carefully selected examples or favorable experimental settings. I am broadly flexible in research direction and committed to driving projects efficiently from idea development to paper submission.

Education

University of Texas at Dallas

PH.D. IN COMPUTER SCIENCE

- Advisor: Feng Chen

Richardson, United States

Aug. 2024 - Present

University of Virginia

M.S. IN COMPUTER SCIENCE, GPA: 3.936/4.0

Charlottesville, United States

Aug. 2022 - Dec. 2023

Xidian University

B.S. IN COMPUTER SCIENCE AND TECHNOLOGY, GPA: 3.7/4.0

Xi'an, China

Sep. 2018 - Jun. 2022

Selected Publication and Preprint

Conditional Contrastive Confidence-Based Uncertainty Quantification for LLMs

Yifan Zhang, Yuzhe Ou, Kangshuo Li, Feng Chen

Accepted, PAKDD 2026

- Proposed C^3 , a sampling-free uncertainty quantification method for LLMs that estimates response-level uncertainty through contrastive token-distribution shifts under learned system-instruction perturbations.
- Designed a gradient-guided search objective to automatically discover perturbations that separate confident and hallucination-prone response regions, requiring only one additional forward pass.
- Demonstrated strong performance across five LLM backbones and two QA benchmarks, achieving the highest AUROC/AUPR in 8/10 and 9/10 head-to-head comparisons against state-of-the-art baselines while reducing inference cost by over 97% ($36\times$ faster than SelfCheckGPT, $113\times$ faster than Semantic Entropy).
- Contribution: Led the project end-to-end, including research conception, method development, experimental design, implementation, result analysis, and initial manuscript writing.

Small Updates, Big Doubts: Does Parameter-Efficient Fine-tuning Enhance Uncertainty Awareness for Large Language Models?

Xu Hu*, Yifan Zhang*, Songtao Wei, Qiannan Li, Chen Zhao, Bingzhe Li, Feng Chen

Under review at TMLR

- Conducted a controlled empirical study on how parameter-efficient fine-tuning (PEFT) reshapes uncertainty-aware hallucination detection in fact-seeking question answering. The study finds that PEFT improves answer accuracy by only 1–3% yet boosts detection AUROC by up to 8.7%, showing that PEFT primarily acts as an uncertainty reshapener—shifting incorrect answers away from overconfident regimes to make them more detectable—rather than substantially expanding factual knowledge.
- Contribution: Conducted all uncertainty-related experiments and contributed substantially to manuscript writing.

Listed above are selected first-author and co-first-author manuscripts. I have also been involved in additional collaborative research projects, several of which are currently under review or in preparation.

Research Experience

AI Safety Lab, UT Dallas

Richardson, United States

RESEARCH ASSISTANT, SUPERVISOR: PROF. FENG CHEN

Aug. 2024 – Present

- **Conditional Contrastive Confidence (C^3) for LLM Uncertainty Quantification**
 - Independently identified the research direction within two months of joining the lab; drafted a brief research outline exploring multiple candidate methodological directions, approved by advisor for execution.
 - Iteratively developed and refined the method, experimental design, and narrative over five months of continuous experimentation (Oct. 2024 – Mar. 2025).
 - Completed the first manuscript and submitted in Mar. 2025; accepted at PAKDD 2026.
- **Parameter-Efficient Fine-tuning and Uncertainty-Aware Hallucination Detection in LLMs**
 - Joined the project in Sep. 2025; took full ownership of all uncertainty-related experiments, including evaluation of seven black-box and white-box hallucination detectors.
 - Co-authored the manuscript alongside experimental work.
 - Completed the first manuscript and submitted in Jan. 2026; currently under review.

University of Virginia

Charlottesville, United States

MASTER'S STUDENT, SUPERVISOR: PROF. AIDONG ZHANG

Feb. 2023 – Dec. 2023

- **Gradient Noise Distribution and Gradient Clipping in Federated Learning**
 - Demonstrated that gradient clipping improves robustness to data heterogeneity for pre-trained models but degrades performance for models trained from scratch, attributing the difference to Gaussian vs. heavy-tailed gradient noise distributions; validated across multiple architectures (ViT, ResNet), heterogeneity levels, participation regimes, and FL algorithms (FedAvg, FedNova, FedProx, Scaffold).

Skill Set

- **Programming Language:** Proficient in Python, C, SQL; Familiar with C++, C#, Java, HTML
- **Language Skills:** Chinese (Native); English (Fluent)
- **Core courses:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Convex Optimization, Geometry of Data, Design and Analysis of Algorithms, Computer Architecture, Digital Signal Processing, Software Security, Data Structures, Probability and Mathematical Statistics